My Learning Draft

Labix

February 9, 2026

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1 More on Simplicial Homology

1.1 Relation to Singular Homology

Theorem 1.1.1 Let X be a topological space endowed with a Δ-complex structure $(T, |T| \cong X)$. The induced map $H_n^{\Delta}(X) \to H_n(X)$ by the map $s : \Delta^n \to X$ for each $s \in T^n$ is an isomorphism.

Proof Firstly, every n-simplex $s \in T$ induces a canonical continuous map $s : \Delta^n \to X$. This extends to a homomorphism $\Delta_n(T) \to C_n(X)$ and so to a chain map and descends to homology.

Denote T^k the Δ-sets of simplices in T of dimension at most k. Define $\Delta_\bullet(T^k, T^{k-1}) = \Delta_\bullet(T^k)/\Delta_\bullet(T^{k-1})$ and accordingly, $H_n(\Delta_\bullet(T^k, T^{k-1}))$. Then the short exact sequence of chain complexes

$$0 \longrightarrow \Delta_\bullet(T^{k-1}) \longrightarrow \Delta_\bullet(T^k) \longrightarrow \Delta_\bullet(T^k, T^{k-1}) \longrightarrow 0$$

Together with the inclusion maps and by naturality in theorem 1.4.3, give the following:

$$\begin{tikzcd}
\cdots \arrow[r] & H_{n+1}(T^k, T^{k-1}) \arrow[r] \arrow[d] & H_n(T^{k-1}) \arrow[r] \arrow[d] & H_n(T^{k-1}) \arrow[r] \arrow[d] & H_n(T^k, T^{k-1}) \arrow[r] \arrow[d] & H_{n-1}(T^{k-1}) \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & H_{n+1}(|T^k|, |T^{k-1}|) \arrow[r] & H_n(|T^{k-1}|) \arrow[r] & H_n(|T^{k-1}|) \arrow[r] & H_n(|T^k|, |T^{k-1}|) \arrow[r] & H_{n-1}(|T^{k-1}|) \arrow[r] & \cdots
\end{tikzcd}$$

When $k = 0$, $\left|T^0\right|$ is a discrete topological space on the set T^0, and the map $H_n(T^0) \to H_n(\left|T^0\right|)$ is an isomorphism. By induction and the five lemma, the middle map is an isomorphism provided that the first and fourth maps are isomorphisms.

Note that $\Delta_\bullet(T^{k-1})$ is a chain complex in degrees $k-1$, $k-2$, down to 0. And $\Delta_\bullet(T^k)$ is the same chain complex with an additional term $\mathbb{Z}T_k$ in degree k. It follows that $\Delta_\bullet(T^k, T^{k-1})$ is the chain complex with $\mathbb{Z}T^k$ concentrated in degree k. In particular, we have that

$$H_n(T^k, T^{k-1}) = \begin{cases} \mathbb{Z}T^k & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

By construction of the geometric realization, we have a homeomorphism

$$\frac{|T^k|}{|T^{k-1}|} \cong \frac{\Delta^k \times T^k}{\partial\Delta^k \times T^k} \cong \bigvee_{T_k} \frac{\Delta^k}{\partial\Delta^k}$$

Using lemma 6.2.4 and lemma 6.2.5, we have that

$$H_n(\left|T^k\right|, \left|T^{k-1}\right|) \cong \widetilde{H_n}(\left|T^k\right|/\left|T^{k-1}\right|) \cong \begin{cases} \mathbb{Z}T_k & \text{if } n = k \\ 0 & \text{otherwisev} \end{cases}$$

with generators corresponding to elements $s \in T^k$ via the relative cycles $s : \Delta^k \to \left|T^k\right|$. It follows that an isomorphism occurs in the first and fourth position of the long exact sequence of relative homology groups.

If $T = T^n$ for some $n \in \mathbb{N}$, we are done.

Continuing the proof, let $z \in Z_n(T)$ be an n-cycle whose image in $H_n(|T|)$ vanishes. Then there exists $\tau \in C_{n+1}(|T|)$ with $\partial\tau = z$. Now since $|T|$ is the geometric realization, we must have every compact subspace of $|T|$ lying in some $\left|T^k\right|$. Hence $\tau \in C_{n+1}(\left|T^k\right|)$ for some $k > n$. We deduce that z maps to 0 in $H_n(\left|T^k\right|)$. By the previous argument, we have taht $z = 0 \in H_n(T^k) = H_n(T)$. This shows injectivity of $H_n(T) \to H_n(|T|)$.

Similarly, let $\sigma \in Z_n(T)$ be an n-cycle. We have seen that $\sigma \in Z_n(\left|T^k\right|)$ for some $k > n$ and by the argument above, $[\sigma]$ comes from $H_n(T^k) = H_n(T)$. This shows surjectivity of $H_n(T) \to H_n(|T|)$ and so we are done. ∎

We can finally show that simplicial homology is independent of the choice of Δ-complex structure.

Corollary 1.1.2 The simplicial homology $H_\bullet^\Delta(X)$ depends on X only and not on the Δ-complex structure.

Proof Indeed the geometric realization of any two Δ-complex structure is isomorphic to the same singular homology group. ■

Corollary 1.1.3 Suppose X has a Δ-complex structure with simplicies in dimension $\leq k$ only. Then $H_n(X) = 0$ for all $n > k$.

Proof Direct since singular homology coincides with simplicial homology. ■

1.2 Simplicial Cohomology

Definition 1.2.1 (Simplicial Cohomology Groups)
Let X be a Δ-complex. Let R be a ring. Let M be an R-module. Define the nth simplicial cohomology group of X

$$H_\Delta^n(X; M) = H^n(\mathrm{Hom}(\Delta_\bullet(X), M))$$

to be the nth cohomology group of the dual of the simplicial complex $\Delta_\bullet(X)$ of X.

1.3

Proposition 1.3.1
Let S be a simplicial complex. Let $\sigma \in S$. Let $p \in |\sigma|^\circ$. Then we have

$$H_{\text{sing}}^k(|S|, |S| \setminus p) \cong \widetilde{H}_\Delta^{k-|\sigma|}(\mathrm{lk}(\sigma))$$

2 Polyhedrons and Polytopes

2.1 Convex Geometry

Definition 2.1.1 (Convex Sets)
Let $C \subseteq \mathbb{R}^n$. We say that C is convex if for all $x, y \in C$ and $t \in [0,1]$, we have $(1-t)x + ty \in C$.

Definition 2.1.2 (Convex Hull)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the convex hull of S to be

$$\text{Conv}(S) = \left\{ \sum_{k=1}^{n} \lambda_k x_k \;\middle|\; x_k \in S \text{ and } \sum_{k=1}^{n} \lambda_k = 1 \right\}$$

Proposition 2.1.3
Let $S \subseteq \mathbb{R}^n$ be a subset. Then the following are sets are equal.
- The convex hull $\text{Conv}(S)$.
- $\bigcap_{C \subseteq \mathbb{R}^n \text{ is convex}} C$.
- The smallest convex set containing S.

2.2 Polyhedrons and its Properties

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in $\mathbb{R}^n$ since Euclidean spaces maybe identified with $\mathbb{R}^n$ canonically.

Definition 2.2.1 (Halfspaces)
A halfspace of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$\{x \in \mathbb{R}^n \mid \langle a, x \rangle + b \geq 0\}$$

for some $a, b \in \mathbb{R}^n$.

Definition 2.2.2 (Polyhedrons)
A polyhedron in $\mathbb{R}^n$ is an intersection of finitely many halfspaces.

Definition 2.2.3 (Polytopes)
A polytope in $\mathbb{R}^n$ is a polyhedron that is bounded.

Proposition 2.2.4
Let $P \subseteq \mathbb{R}^n$ be a subset. Then P is a polytope if and only if there exists a finite subset $S \subseteq \mathbb{R}^n$ such that $P = \text{Conv}(S)$.

Definition 2.2.5 (Dimension)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dimension of P to be the dimension of the smallest affine subspace containing P.

Definition 2.2.6 (Affinely Isomorphic)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are affinely isomorphic if there exists an affine transformation $f : \mathbb{R}^{n_1} \to \mathbb{R}^{n_2}$ such that $f|_{P_1}$ is a bijection.

Note that f itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

2.3 The Faces of a Polyhedron

Definition 2.3.1 (Supporting Hyperplane)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $H \subseteq \mathbb{R}^n$ be a hyperplane. We say that H is a supporting hyperplane for P if the following are true.

- $P \cap H \neq \emptyset$.
- P lies entirely in one of the halfspaces defined by H.

Definition 2.3.2 (Faces)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. A face of P is a subset of P defined by

$$F = P \cap H$$

where H is a supporting hyperplane for P.

Lemma 2.3.3
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face. Then F is a polytope.

Definition 2.3.4 (Vertices, Edges and Facets)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face.

- We say that F is a vertex if $\dim(F) = 0$.
- We say that F is an edge if $\dim(F) = 1$.
- We say that F is a facet if $\dim(F) = \dim(P) - 1$.

Definition 2.3.5 (The Face Poset)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Define the face poset $\mathcal{F}(P)$ of P to be the set of faces of P together with inclusion.

Definition 2.3.6 (The Face Vector)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Define the face vector $(f_0, \dots, f_{d-1})$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } P$$

for $0 \leq i \leq d$ and by convention, $f_{-1} = f_d = 1$.

Lemma 2.3.7 (Euler's Formula)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Then we have

$$\sum_{i=0}^{d-1} (-1)^i f_i = 1 + (-1)^d$$

Definition 2.3.8 (Combinatorially Equivalent)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are combinatorially equivalent if there exists a inclusion preserving bijection between $\mathcal{F}(P_1)$ and $\mathcal{F}(P_2)$. In this case we write $P_1 \simeq P_2$.

3 Duality in Polytopes

3.1 Polar Sets

Definition 3.1.1 (Polar Sets)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the polar set of S to be

$$S^* = \{f \in (\mathbb{R}^n)^* \mid f(x) \leq 1 \text{ for all } x \in S\}$$

Proposition 3.1.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. If $0 \in P$, then P^* is a polytope.

Example 3.1.3
The polar set of the cube $C^3 = \{(a, b, c) \in \mathbb{R}^3 \mid -1 \leq a, b, c \leq 1\}$ is the octahedron

$$(C^3)^* = \{(x, y, z) \in (\mathbb{R}^3)^* \mid \pm x + \pm y + \pm z \leq 1\}$$

$P = (P^*)^*$.

$\mathcal{F}(P)^{\text{op}} \cong \mathcal{F}(P^*)$.

3.2 Simplicial Polytopes

Definition 3.2.1 (Simplicial Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simplicial polytope if all its facets are simplicies.

Lemma 3.2.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. Then ∂P is a simplicial complex.

Proposition 3.2.3 (The Dehn-Sommerville Equation)
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope of dimension d. Then the following are true:

$$\sum_{j=k}^{d-1} (-1)^j \binom{j+1}{k+1} f_j = (-1)^{d-1} f_k$$

for $-1 \leq k \leq d-2$.

In particular, ∂P is a simplicial complex. The notion of face vector coincides, and they enjoy both the Dehn-Sommerville equation and its relation with the h-vector.

3.3 Simple and Simplicial Polytopes

Definition 3.3.1 (Simple Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simple polytope if for all vertices $v \in P$, there exists n edges $e_1, \dots, e_n \subseteq P$ such that $v \in \partial e_i$ for all i.

Lemma 3.3.2
Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then the Cartesian product $P_1 \times P_2$ is a simple polytope.

3.4 The Duality Between Simple and Simplicial Polytopes

Proposition 3.4.1
Let $P \subseteq \mathbb{R}^n$. Then P is simple if and only P^* is simplicial.

Duality of d-dimensional faces with $(n-d)$-dimensional faces.

Proposition 3.4.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. If P is both simple and simplicial, then P belongs to one of the following types of polytopes.
- $n = 2$ (and so P is a polygon).
- P is a simplex.

3.5 Nerve Complex

Definition 3.5.1 (The Nerve Complex)
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Define the nerve complex of P to be the simplicial complex

$$\mathcal{K}_P = \partial(P^*)$$

Lemma 3.5.2
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Then $\mathcal{K}_P$ is a triangulation of S^{n-1}.

Lemma 3.5.3
Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then we have

$$\mathcal{K}_{P_1 \times P_2} \cong \mathcal{K}_{P_1} * \mathcal{K}_{P_2}$$

4 Piecewise Linear Topology

4.1 Relation between Polyhedra and Simplicial Complexes

Definition 4.1.1 (Locally Finite Collections)
Let X be a space. Let $P \subseteq \mathcal{P}(X)$ be a collection of subsets. We say that P is locally finite if for each $p \in X$, there exists a neighbourhood $U \subseteq X$ of p such that the set

$$\{V \in P \mid U \cap V \neq \emptyset\}$$

is finite.

Proposition 4.1.2
Let P be a polyhedra. Then P is a locally finite union of simplexes.

Thus if P is a polytope, then P has the underlying structure of a simplicial complex.

4.2 Piecewise Linear Maps

Definition 4.2.1 (Piecewise Linear Maps between Polyhedra)
Let P, Q be polyhedra. Let $f : P \to Q$ be a function. We say that f is piecewise linear if there exists a locally finite decomposition into polyhedra $P = \bigcup_{\alpha \in I} P_\alpha$ such that for each $\alpha \in I$, the restriction $f|_{P_\alpha}$ is affine linear.

Proposition 4.2.2
Let P, Q be polyhedra. Let $f : P \to Q$ be a function. Then the following are equivalent.
- f is piecewise linear.
- For each $p \in P$, there exists $L \subseteq P$ such that $f(\lambda p + \mu x) = \lambda f(p) + \mu f(x)$ for all $\lambda, \mu \geq 0$ and $\lambda + \mu = 1$ and $x \in L$.
- $\Gamma(f) = \{(x, f(x)) \mid x \in P\}$ is a polyhedron.
- There exists locally finite decompositions of P and Q into simplexes such that f sends simplexes to simplexes and f restricted to each simplex is affine linear.

Definition 4.2.3 (PL Homeomorphism)
Let P, Q be polyhedra. Let $f : P \to Q$ be a piecewise linear function. We say that f is a PL homeomorphism if there exists a piecewise linear function $g : Q \to P$ such that

$$g \circ f = \mathrm{id}_P \quad \text{and} \quad f \circ g = \mathrm{id}_Q$$

Lemma 4.2.4
Let P, Q be polyhedra. Let $f : P \to Q$ be a piecewise linear function. If f is a homeomorphism, then f is a piecewise linear homeomorphism.

Motivated by the equivalent definitions of piecewise linear maps between polyhedra, we define piecewise linear maps between spaces in general.

Definition 4.2.5 (Piecewise Linear Maps in General)
Let X, Y be spaces. Let $f : X \to Y$ be a function. We say that f is piecewise linear if there exists locally finite triangulations of X and Y such that f restricted to each simplex is affine linear.

Lemma 4.2.6
Let S, T be simplicial complexes. Let $f : V(S) \to V(T)$ be a simplicial map. Then $|f| : |S| \to |T|$ is piecewise linear.

Proposition 4.2.7
Let S,T be simplicial complexes. Let $f:|S|\to|T|$ be a function. Then f is piecewise linear if and only if there exists subdivisions S' of S and T' of T such that $f:|S'|\to|T'|$ is simplicial.

4.3 Piecewise Linear Manifolds

Proposition 4.3.1
The set
$$\Gamma=\{f:U\to V\mid U,V\subseteq\mathbb{R}^n\text{ are open },f\text{ is a PL homeomorphism }\}$$
defines a pseudogroup of transformations on $\mathbb{R}^n$.

Definition 4.3.2 (Piecewise Linear Manifolds)
A piecewise linear manifold is a manifold whose pseudogroup of transformations is given by Γ.

4.4 Triangulated Manifolds

Definition 4.4.1 (Triangulated Manifolds)
Let M be a topological manifold. We say that M is a triangulated manifold if M is triangulated as a topological space.

Proposition 4.4.2
Let X be a piecewise linear manifold. Then X is a triangulated manifold.

Proposition 4.4.3
Let M be a triangulated manifold triangulated by the simplicial complex K. Then the following are equivalent.
- M is a PL manifold.
- For every $\emptyset\neq\sigma\in K$, we have that lk(σ) is PL homeomorphic to $\left|\Delta^d\right|$ where $d=\dim(M)-|\sigma|$.
- For every $p\in V(K)$, we have that st(p) is PL homeomorphic to $\left|\Delta^{\dim(M)}\right|$.

The Hauptvermutung asks whether any two homeomorphic triangulated spaces are PL homemorphic (i.e. if S,T are simplicial complexes and $|S|$ is homeomorphic to $|T|$, is there a PL homeomorphism between $|S|$ and $|T|$?). This is shown to be False by Milnor using Reidemeister torsion.

The manifold Hauptvermutung asks whether any two homeomorphic triangulated manifolds are PL homemorphic. This is also show to be false using an obstruction in cohomology.

Proposition 4.4.4 Every finite CW complex is homotopy equivalent to a topological manifold.

4.5 Pseudo-Manifolds

Definition 4.5.1 (Pseudo-Manifolds From Simplicial Complexes)
Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is a pseudo-manifold of dimension n if the following are true.
- All maximal simplicies of $\mathcal{K}$ are of dimension n.
- Each $(n-1)$-simplex of $\mathcal{K}$ is the face of two n-simplicies of $\mathcal{K}$.
- For any two n-simplicies $\sigma,\sigma'\in\mathcal{K}$, there exists n-simplicies $\sigma=\sigma_0,\dots,\sigma_k=\sigma'$ such that $\sigma_i\cap\sigma_{i+1}$ is an $(n-1)$-simplex for $0\leq i\leq k-1$.

Definition 4.5.2 (Pseudo-Manifolds in General)
Let X be a space. We say that X is a pseudo-manifold if X is triangulated by a simplicial complex

$\mathcal{K}$ that is a pseudo-manifold.

Lemma 4.5.3
Let X be a pseudo-manifold of dimension n. Then $H_n(X) \cong \mathbb{Z}$ or 0.

Lemma 4.5.4
Let $\mathcal{K}$ be a triangulation of S^n. Then the following are true.

- $\mathcal{K}$ is a pseudo-manifold.
- If $\mathcal{K}$ is finite, then the generator of $H_n(\mathcal{K})$ is represented by the sum of n-simplicies of $\mathcal{K}$.

5 Cones and Fans

5.1 Cones and its Properties

Definition 5.1.1 (Cones)
Let $C \subseteq \mathbb{R}^n$. We say that C is a cone if $x \in C$ implies that $sx \in C$ for all $s \geq 0$.

Definition 5.1.2 (Polyhedral Cones)
A cone in $\mathbb{R}^n$ is a subset $C \subseteq \mathbb{R}^n$ such that if $c_1, \dots, c_n \in C$, then $\sum_{i=1}^n \lambda_i c_i \in C$ for all $\lambda_1, \dots, \lambda_n \geq 0$.

Definition 5.1.3 (Conical Hull)
Let $S \subseteq \mathbb{R}^n$ be finite. Define the conical hull of S to be

$$\text{Cone}(S) = \left\{ \sum_{s \in S} \lambda_s s \;\middle|\; \lambda_s \geq 0 \right\}$$

Proposition 5.1.4
Let $C \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.
- C is a polyhedral cone.
- There exists a finite subset $S \subseteq \mathbb{R}^n$ such that $C = \text{Cone}(S)$.
- C is the intersection of finitely many halfspaces containing 0 in their boundary.

Therefore cones are also polyhedrons.

Definition 5.1.5 (Dimension of a Cone)
Let $C \subseteq \mathbb{R}^n$ be a cone. Define the dimension of C to be the dimension of the smallest affine subspace containing C.

Definition 5.1.6 (Rational Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is rational if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that $C = \text{Cone}(S)$.

Definition 5.1.7 (Regular Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is regular if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that the following are true.
- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{Z}$-linearly independent.

Definition 5.1.8 (Simplicial Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is simplicial if there exists a finite set $S \subseteq \mathbb{R}^n$ such that the following are true.
- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{R}$-linearly independent.

5.2 The Dual Cone

Definition 5.2.1 (The Dual of a Cone)
Let $C \in \mathbb{R}^n$ be a cone. Define the dual of C to be

$$C^\vee = \{x \in \mathbb{R}^n \mid \langle x, c \rangle \geq 0 \text{ for all } c \in C\}$$

Definition 5.2.2 (Associated Hyperplane and Halfspace of a Linear Functional)
Let $\phi \in (\mathbb{R}^n)^*$. Define the associated hyperplane of ϕ to be

$$H_\phi = \{x \in \mathbb{R}^n \mid \phi(x) = 0\}$$

Define the associated halfspace of ϕ to be

$$H_\phi^+ = \{x \in \mathbb{R}^n \mid \phi(x) \geq 0\}$$

Lemma 5.2.3
Let $C \in \mathbb{R}^n$ be a cone. Let H be a hyperplane in $\mathbb{R}^n$. Then H is a supporting hyperplane of C if and only if there exists $\phi \in C^\vee$ such that $H = H_\phi$.

Lemma 5.2.4
Let $C \in \mathbb{R}^n$ be a polyhedral cone. Then the following are true.
- $C^\vee$ is a polyhedral cone.
- $(C^\vee)^\vee = C$.
- $C = \bigcap_{i=1}^n H_{\phi_i}$ for $\phi_i \in (\mathbb{R}^n)^*$ if and only if $C^\vee = \text{Cone}(\phi_1, \dots, \phi_n)$.
- If C is strongly convex, then $C^\vee$ is strongly convex.

Lemma 5.2.5 (Separation Lemma)
Let $C_1, C_2 \subseteq \mathbb{R}^n$ be cones such that $F = C_1 \cap C_2$ is a face of both C_1 and C_2. Then there exists $\phi \in C_1^\vee \cap (-C_2)^\vee$ such that

$$F = C_1 \cap H_\phi = C_2 \cap H_\phi$$

Definition 5.2.6 (Strongly Convex Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is strongly convex if C does not contain a line.

Lemma 5.2.7
Let $C \subseteq \mathbb{R}^n$ be a cone. Then the following are equivalent.
- C is strongly convex.
- $\{0\}$ is a face of C.
- $C \cap (-C) = \{0\}$.
- $\dim(C^\vee) = n$.

5.3 Fans and its Properties

Definition 5.3.1 (Fans)
Let $n \in \mathbb{N}^+$. A fan in $\mathbb{R}^n$ is a finite set of cones Σ such that the following are true.
- Each $C \in \Sigma$ is a strongly convex rational polyhedral cone.
- For each cone $C \in \Sigma$ and each face F of C, we have $F \in \Sigma$.
- For cones $C_1, C_2 \in \Sigma$, we have $C_1 \cap C_2$ are faces of C_1 and C_2.

We write $\Sigma(d) = \{C \in \Sigma \mid \dim(C) = d\}$ for the d-dimensional cones in the fan.

Definition 5.3.2 (The Support of a Fan)
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Define the support of Σ to be

$$|\Sigma| = \cup_{C \in \Sigma} C$$

Definition 5.3.3 (Types of Fans)

Definition 5.3.4 (Complete Fans)
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. We say that Σ is complete if $|\Sigma| = \mathbb{R}^n$.

6 Some Results on the Classification of Polytopes

Let P be a simplicial polytope of dimension 4. Since P is homeomorphic to D^4, the following classification problems are equivalent.

- Classification of simplicial polytopes up to uniqueness of the face vector.
- Classification of triangulations of the 3-sphere up to uniqueness of the face vector of the simplicial complex.

(More subtle, this is not true for higher dimensions: see https://math.mit.edu/ rstan/pubs/pubfiles/54.pdf)

Of course, two polytopes that are not combinatorially equivalent can still have the same face vector. Let (f_0, f_1, f_2, f_3) be the face vector of P. A few is known in this particular case.

- Since P is simplicial, we have the bound $f_i \leq \binom{f_0}{i+1}$ from here
- The Euler characteristic of the sphere tells us that $f_0 - f_1 + f_2 - f_3 = 0$.
- shows that $f_1 \geq 4f_0 - 10$.
- The same paper tells us that $f_2 = 2f_1 - 2f_0$ and $f_3 = f_1 - f_0$ (also this paper:
- For every pair of (f_0, f_1) such that $f_0 \geq 5$ and $\frac{f_0(f_0-1)}{2} \geq f_1 \geq 4f_0 - 10$, there exists a triangulation of the 3-sphere whose face vectors are $(f_0, f_1, 2f_1 - 2f_0, f_1 - f_0)$.

Recalling that every d dimensional polytope must have $d + 1$ vertices,we start enumerating the possible cases with $f_0 = 5$.

- By the above, we must have $f_1 \geq 10$, and there is guaranteed to be a triangulation if $f_1 \leq 10$. So $(5, 10, 10, 5)$ is one such face vector. By the upper bound of the face vector, this is the only possible face vector.

6.1

Theorem 6.1.1
The following are true regarding the classification of polytopes up to combinatorial type.
- The number of polytopes of dimension d with $d + 1$ vertices is 1.
- The number of polytopes of dimension d with $d + 1$ vertices is $[1/4d^2]$.

6.2

Definition 6.2.1 (Polytopal Spheres)
Let $\mathcal{K}$ be a simplicial complex that is the triangulation of a sphere. We say that the triangulation is polytopal if $\mathcal{K}$ is the boundary of a simplicial polytope.

Convex polytopes p.424

$$\begin{matrix}\text{Polytopal} \\ \text{Spheres}\end{matrix} \subseteq \begin{matrix}\text{PL} \\ \text{Spheres}\end{matrix} \subseteq \begin{matrix}\text{Triangulated} \\ \text{Spheres}\end{matrix}$$

Let d be the dimension of the sphere (hence the underlying polytope has dimension $d + 1$). If $d = 2$, all inclusions are equalities. If the simplicial complex of the triangulation has at most $d + 4$ vertices, all inclusions are equalities.

The smallest non-trivial case is then $d = 3$ such that the polytope has dimension 4 and 8 vertices. The triangulations are classified by "David Barnette. The triangulations of the 3-sphere with up to 8 vertices" where there are a total of 39 triangulations, of which 37 are polytopal, and the polytopes are given in "BRANKO GRUJNBAUM AND V. P. SREEDHARAN. An Enumeration of Simplicial 4-Polytopes with 8 Vertices".

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7

Pre-req: Commutative Algebra 2, Simplicial Complexes

7.1 Stanley-Reisner Rings

Theorem 7.1.1
Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\begin{array}{c}\text{Abstract Simplicial}\\ \text{Complexes on } \{1,\dots,n\}\end{array} \overset{1:1}{\leftrightarrow} \begin{array}{c}\text{Squarefree monomial}\\ \text{ideals of } R[x_1,\dots,x_n]\end{array}$$

given by sending a simplicial set S to the ideal $\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$.

Definition 7.1.2 (Stanley-Reisner Rings)
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of S to be

$$R[S] = \frac{k[x_1,\dots,x_n]}{\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle}$$

The ideal $I_S = \left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$ is called the Stanley-Reisner ideal of S.

Example 7.1.3
Let R be a commutative ring. The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$R[\Delta^n] = k[x_1,\dots,x_n]$$

Example 7.1.4
Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$R[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 7.1.5
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_S = \bigcap_{J\in S} \langle x_i \mid i \notin J \rangle$$

Maps of simplicial sets -> Maps of k-algebra.

Proposition 7.1.6
Let R be a commutative ring. Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$R[\mathcal{K}_1 * \mathcal{K}_2] \cong R[\mathcal{K}_1] \otimes_R R[\mathcal{K}_2]$$

Proposition 7.1.7
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a flag complex, then $R[\mathcal{K}]$ is a Koszul algebra.

7.2 The Hilbert Series of Stanley-Reisner Rings

Theorem 7.2.1 (Stanley)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S. Let $(h_0, \dots, h_n)$ be the h-vector of S. Then the Hilbert series of $k[S]$ is given by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1} \left(\frac{t^2}{1-t^2} \right)^k = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} h_i t^{2i}$$

Example 7.2.2 | Let Δ^n be the standard n-simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

8 Moment-Angle Complexes

Pre req: Equivariant (co)homology, homological algebra, PL manifolds?

8.1 Basic Definitions

Proposition 8.1.1
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Suppose that $(S^1)^n \subseteq \mathbb{C}^n$ act on $\mathbb{C}^n$ by left multiplication. Then μ descends to a homeomorphism of orbit spaces

$$\frac{\mathbb{C}^n}{(S^1)^n} \cong \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i\} \subseteq \mathbb{R}^n$$

Lemma 8.1.2
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Then the above homeomorphism induces a homeomorphism

$$\frac{D^n}{(S^1)^n} \cong [0, 1]^n$$

Recall the construction of the cubical embedding $i_c : \Delta^n \to [0, 1]^{n+1}$.

1. For $1 \leq k \leq n$, the barycenter of the k-dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_n) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \to [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n, $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Since every finite abstract simplicial complex can be realized as a subcomplex of Δ^n for large n, we also get the cubical embedding of a finite abstract simplicial complex, which is denoted by $\text{cc}(\mathbb{K})$.

Definition 8.1.3 (Moment-Angle Complex)
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Define the moment-angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = \mu^{-1}(\text{cc}(\mathcal{K})) \subseteq D^n$$

Equivalently, it is the pullback of the inclusion map $\mathcal{Z}_{\mathcal{K}} = \text{cc}(\mathcal{K}) \to [0, 1]^n$ and the quotient map $\mu : D^n \to [0, 1]^n$.

Example 8.1.4
Let $n \in \mathbb{N}$. Then the following are true.

- $\mathcal{Z}_{\emptyset} = (S^1)^n$.
- $\mathcal{Z}_{\Delta^n} = D^{n+1}$.

Lemma 8.1.5
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. The following are true regarding the moment-angle complex of $\mathcal{K}$.

- $\mathcal{Z}_{\mathcal{K}}$ is a $(S^1)^n$-invariant subspace.
- The homeomorphism $\frac{D^n}{(S^1)^n} \cong [0, 1]^n$ induced by the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$ descends to a homeomorphism

$$\frac{\mathcal{Z}_{\mathcal{K}}}{(S^1)^n} \cong \text{cc}(\mathcal{K})$$

Recall that $\text{cc}(\mathcal{K})$ is equal to the union of a finite number of faces of $[0, 1]^n$. Also recall the notation

$$C_I = \{(x_1, \dots, x_n) \in [0, 1]^n \mid x_i = 0 \text{ if } i \in I\}$$

for a face of the cube $[0, 1]^n$ that contains the origin and $I \subseteq \{1, \dots, n\}$.

Lemma 8.1.6
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Suppose that $\text{cc}(\mathcal{K}) = \bigcup_{I \in \mathcal{K}} C_I$. Then there is a homeomorphism

$$\mathcal{Z}_\mathcal{K} = \bigcup_{I \in \mathcal{K}} \mu^{-1}(C_I) = \bigcup_{I \in \mathcal{K}} \{(x_1, \dots, x_n) \in D^n \mid |z_j|^2 = 1 \text{ for all } j \in \{1, \dots, n\} \setminus I\}$$

where μ is the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$.

For $I \subseteq \{1, \dots, n\}$, the condition that $|z_j|^2 = 1$ for $j \in \{1, \dots, n\} \setminus I$ means that for those coordinates in D^n, they only consist of the boundary which is S^1. Hence we may think of $\mathcal{Z}_\mathcal{K}$ as the union of products

$$\mathcal{Z}_\mathcal{K} = \bigcup_{I \in \mathcal{K}} \left(\prod_{i \in I} D^2 \times \prod_{i \in \{1, \dots, n\} \setminus I} S^1 \right)$$

But note that this product is ambiguous on which coordinates should be S^1 and which should be D^2 (because we have a chosen embedding into $\mathbb{C}^n$ although swapping the order of the products give homeomorphic spaces). Moreover, because we are taking the union over all $I \in \mathcal{K}$, it suffices to consider the maximal simplicies in $\mathcal{K}$.

Example 8.1.7
Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.
- The moment angle complex of $\partial \Delta^n$ is given by

$$\mathcal{Z}_{\partial \Delta^n} \cong S^{2n+1}$$

- The moment angle complex of the n-gon is given by

Proof
It suffices to consider the maximal simplicies in $\partial \Delta^n$. They are of the form $I \subseteq \{1, \dots, n+1\}$ for $|I| = n$. Hence we have

$$\begin{aligned}
\mathcal{Z}_{\partial \Delta^n} &= (D^2 \times \cdots \times D^2 \times S^1) \cup \cdots \cup (S^1 \times D^2 \times \cdots \times D^2) \\
&= \partial(D^2 \times \cdots \times D^2) \\
&\cong \partial D^{2n+2} \\
&\cong S^{2n+1}
\end{aligned}$$

For the moment angle complex of the n-gon, we note that the simplicial complex is given by 1-simplicies of the form $\{i, i+1\}$ for $1 \leq i \leq n$ together with $\{1, n+1\}$. Hence

$$\mathcal{Z}_{n\text{-gon}} = (D^2 \times D^2 \times S^1 \times \cdots \times S^1) \cup \cdots \cup (S^1 \times \cdots \times S^1 \times D^2 \times D^2) \cup (D^2 \times S^1 \times \cdots \times S^1 \times D^2)$$

■

Proposition 8.1.8
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then $\mathcal{Z}_\mathcal{K}$ is a closed topological manifold of dimension $n + m$.

8.2 Polyhedral Products

Definition 8.2.1 (Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Define the polyhedral product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be

$$(\mathbf{X},\mathbf{A})^{\mathcal{K}} = \bigcup_{I\in\mathcal{K}}\left\{(x_1,\dots,x_n)\in\prod_{i=1}^n X_i \mid x_i\in A_i \text{ for } i\notin I\right\} = \bigcup_{I\in\mathcal{K}}\left(\prod_{i\in I}X_i\times\prod_{i\notin I}A_i\right)$$

8.3 Algebraic Invariants of the Moment Angle Complex

Proposition 8.3.1
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There is an isomorphism of rings
$$H^*((\mathbb{CP}^\infty,*)^{\mathcal{K}};R)\cong R[\mathcal{K}]$$
- The inclusion $(\mathbb{CP}^\infty,*)^{\mathcal{K}}\hookrightarrow(\mathbb{CP}^\infty,*)^n$ induces the quotient map
$$H^*((\mathbb{CP}^\infty)^m;R)\cong R[v_1,\dots,v_n]\to R[\mathcal{K}]\cong H^*((\mathbb{CP}^\infty,*)^{\mathcal{K}};R)$$
(where $v_1,\dots,v_n$ are degree 2 elements).
- The inclusion $i:(\mathbb{CP}^\infty,*)^{\mathcal{K}}\hookrightarrow(\mathbb{CP}^\infty,*)^n$ decomposes into a homotopy equivalence and fibration in the following way:
$$\begin{tikzcd}
(\mathbb{CP}^\infty,*)^{\mathcal{K}} \arrow[rr, hook] \arrow[dr, "\simeq"'] & & (\mathbb{CP}^\infty,*)^n \\
& ET^m\times_{T^m}\mathcal{Z}_{\mathcal{K}} \arrow[ur, "\text{fib.}"'] &
\end{tikzcd}$$
Moreover, the fiber of the fibration is $\mathcal{Z}_{\mathcal{K}}$.
- $\mathcal{Z}_{\mathcal{K}}(BS^1,*)=ET^m\times_{T^m}\mathcal{Z}_{\mathcal{K}}$.
- $H^*(ET^m\times_{T^m}\mathcal{Z}_{\mathcal{K}})=R[\mathcal{K}]$.

Theorem 8.3.2 ((Davis-Januskiewicz))
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the equivariant cohomology ring of $\mathcal{Z}_{\mathcal{K}}$ is given by

$$H^*_{T^m}(\mathcal{Z}_{\mathcal{K}};R)\cong R[\mathcal{K}]$$

(natural in $\mathcal{K}$)

Proposition 8.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true regarding the homotopy groups of $\mathcal{Z}_{\mathcal{K}}$.

- $\mathcal{Z}_{\mathcal{K}}$ is two-connected.
- $\pi_i(\mathcal{Z}_{\mathcal{K}})\cong\pi_i((\mathbb{CP}^\infty,*)^{\mathcal{K}})$ for $i\geq 3$.

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1\in D^2$.
- One 1-cell S^1.
- One 2-cell D^2.

Let $n\in\mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J,I\subseteq\{1,\dots,n\}$ with $J\cap I=\emptyset$, we denote $\chi(J,I)$ as the $(|J|+|I|)$-cell corresponding to the cell with the factor of a 1-cell in the jth position for $j\in J$, the factor of a 0-cell in the ith position for $i\in I$, and the factor of a 0-cell for $k\notin I\cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then $\mathcal{Z}_\mathcal{K}$ is a union of cells in D^{2n}, and so is a cellular subcomplex of D^{2n}. Namely, for each $I\in\mathcal{K}$, we have that $\chi(J,I)\subseteq\mathcal{Z}_\mathcal{K}$ for all $J\subseteq\{1,\dots,n\}\setminus I$.

Q: How does the cellular cochain split into a bicomplex

Definition 8.3.4 (Bigrading in the Cochain Complex)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define a bigrading in $C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ as follows. For each basis element $\chi(J,I)$, define $\deg(\chi(J,I))=(-|J|,2(|I|+|J|))$.

Lemma 8.3.5
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- The $(2j)$th components for each $j\in\mathbb{N}$ defines a cochain complex $C^{-i,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ in i.
- $(C^{-*,*}(\mathcal{Z}_\mathcal{K};\mathbb{Z}),d_{CW},0)$ is a double cochain complex.
- The cellular cochain complex of $\mathcal{Z}_\mathcal{K}$ splits into a direct sum
$$C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\text{Tot}^{\oplus}(C^{-*,*}(\mathcal{Z}_\mathcal{K}))=\bigoplus_{j\in\mathbb{N}}C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$
where
$$C^k(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\bigoplus_{2j-i=k}C^{-i,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

The consequence of the splitting is that cohomology also accquires a splitting
$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\bigoplus_{2j-i=n}H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))$$

Proposition 8.3.6
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There are natural isomorphisms
$$H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))\cong H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}=H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)_{2q}\right)$$
where $\Lambda\left(\bigoplus_{i=1}^n R[\mathcal{K}]u_i\right)$ is the differential graded algebra whose homological grading is given by $\deg(u_i)=-1$, internal grading is given by $\deg(v_i)=2$ and differential is given by $d(u_i)=v_i$ (and $d(v_i)=0$ since we think of the DGA as a DGA over the ring $\mathbb{Z}[\mathcal{K}]$).
- There above natural isomorphisms induces natural isomorphisms
$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong\bigoplus_{2q-p=k}H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}$$
- The above natural isomorphisms induce a natural isomorphism of rings
$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong H_*\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)$$

Note that $H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is not isomorphic to $H_{-n}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)$ despite the suggestive notation. This is a mismatch of algebraic objects since the left is an abelian group while the right hand side is a $\mathbb{Z}[\mathcal{K}]$-module. In order to have an isomorphism of abelian groups we must further decompose each homology $\mathbb{Z}[\mathcal{K}]$-modules further into $\mathbb{Z}$-modules by considering $\mathbb{Z}[\mathcal{K}]$ as a $\mathbb{Z}$-module. But this results in a shift of degrees.

Let $K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ be the Koszul complex over $\mathbb{Z}[v_1,\dots,v_n]$. Notice that the cochain complex defined by $K^n=K_n(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ is precisely our differential graded algebra $\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)$. Then the cohomology of $\mathcal{Z}_\mathcal{K}$ acquires an additional description:

$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=n} \left(H_p\left(K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])\right)\right)_{2q}$$

Finally, since $K_\bullet(v_1,\dots,v_n)$ is a free resolution for $\mathbb{Z}$ as a $\mathbb{Z}[v_1,\dots,v_n]$-module, we know that

$$H_p(K_\bullet(v_1,\dots,v_n;R[\mathcal{K}])) = \text{Tor}_p^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[\mathcal{K}],\mathbb{Z})$$

The Koszul algebra gives the Tor groups the structure of an algebra, and so the cohomology ring acquires the description

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \text{Tor}_*^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[v_1,\dots,v_n],\mathbb{Z})$$

But again note that the nth cohomology group of $\mathcal{Z}_\mathcal{K}$ is not isomorphic to the nth Tor group of $\mathbb{Z}[\mathcal{K}]$ and $\mathbb{Z}$ by virtually the same reason as above.

Proposition 8.3.7
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There are natural isomorphisms of cochain complexes

$$\bigoplus_{J\subseteq\{1,\dots,n\}} C^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z}) \cong C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

  given by

$$\alpha_L \mapsto \varepsilon(L,J)\chi(J\setminus L, L)$$

  where α_L is the cochain basis of Hochster's formula given by the simplex $L \subseteq J$ and

$$\varepsilon(L,J) = \prod_{j\in L}\varepsilon(j,J)$$

  and $\varepsilon(j,J) = (-1)^{r-1}$ if j is in the rth position of J.
- There above natural isomorphisms induces natural isomorphisms

$$H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) \cong \bigoplus_{\substack{J\subseteq\{1,\dots,n\}\\ |J|=j}} \widetilde{H}^{j-i-1}(\mathcal{K}_J;\mathbb{Z})$$

  where by convention we set $\widetilde{H}^{-1}(\mathcal{K}_\emptyset) = \mathbb{Z}$.
- The above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

We are missing a description of the cup product in terms of Hochster's formula. It is given by the following definition.

Definition 8.3.8 (Ring Structure in Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define a ring structure in $\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_\mathcal{J};\mathbb{Z})$ as follows. Let α_L be the cochain basis element in $C^p(\mathcal{K}_I;\mathbb{Z})$ corresponding to the simplex $L\subseteq I$. The ring structure is induced by the map

$$C^p(\mathcal{K}_I;\mathbb{Z})\otimes_\mathbb{Z} C^q(\mathcal{K}_\mathcal{J};\mathbb{Z}) \to C^{p+q+1}(\mathcal{K}_{I\cup J})$$

defined by

$$\alpha_L\otimes\alpha_M \mapsto \begin{cases} \varepsilon(L,I)\varepsilon(M,J)\varepsilon(L\cup M, I\cup J)\alpha_{L\amalg M} & \text{if } I\cap J=\emptyset \\ 0 & \text{otherwise} \end{cases}$$

where $\varepsilon(L,I) = \prod_{j\in L}\varepsilon(j,J)$ and $\varepsilon(j,J) = (-1)^{r-1}$ if j is in the rth position in J.

Proposition 8.3.9
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- The ring structure in Hochster's formula is equal to the relative Kunneth theorem for

$$H^*(|\mathcal{K}_I * \mathcal{K}_J|) = H^*(\Sigma(|\mathcal{K}_I| \wedge |\mathcal{K}_J|)) = \widetilde{H}^{*-1}(|K_I| \times |K_J|, |K_I \vee K_J|)$$

  via the inclusion $\mathcal{K}_{I \amalg J} \hookrightarrow \mathcal{K}_I * \mathcal{K}_J$.
- The natural isomorphism in the above prp induces a natural isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} \widetilde{H}^*(\mathcal{K}_J; \mathbb{Z})$$

While the cohomology groups can be easily computed using Hochster's formula, the ring structure can be seen more easily using the DGA.

Example 8.3.10
Let $\mathcal{K}$ be the standard polygon with vertices given by $\{1, \dots, 5\}$ and edges given by $\{\{1, 2\}, \dots, \{4, 5\}, \{1, 5\}\}$. We compute that

- $H^0(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{J \subseteq \{1,\dots,5\}} \widetilde{H}^{0-|J|-1}(\mathcal{K}_{\mathcal{J}}; \mathbb{Z}) = \widetilde{H}^{-1-|\emptyset|}(\mathcal{K}_\emptyset; \mathbb{Z}) = \mathbb{Z}$.
- $H^1(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^2(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^5(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^6(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = 0$.

Also we have

$$\begin{aligned} H^3(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1,\dots,5\}} \widetilde{H}^{2-|J|}(\mathcal{K}_{\mathcal{J}}; \mathbb{Z}) \\ &= \widetilde{H}^0(\mathcal{K}_{\{1,3\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,4\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,4\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,5\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{3,5\}}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{3\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \\ &= \mathbb{Z}[u_1 v_3] \oplus \mathbb{Z}[u_1 v_4] \oplus \mathbb{Z}[u_2 v_4] \oplus \mathbb{Z}[u_2 v_5] \oplus \mathbb{Z}[u_3 v_5] \end{aligned}$$

$$\begin{aligned} H^4(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1,\dots,5\}} \widetilde{H}^{3-|J|}(\mathcal{K}_{\mathcal{J}}; \mathbb{Z}) \\ &= \widetilde{H}^0(\mathcal{K}_{\{1,2,4\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,3,5\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,3,4\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,4,5\}}; \mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,3,5\}}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{1\}}] \oplus \mathbb{Z}[\alpha_{\{2\}}] \oplus \mathbb{Z}[\alpha_{\{3\}}] \\ &= \mathbb{Z}[-u_1 u_2 v_4] \oplus \mathbb{Z}[-u_2 u_3 v_5] \oplus \mathbb{Z}[-u_3 u_4 v_1] \oplus \mathbb{Z}[-u_4 u_5 v_2] \oplus \mathbb{Z}[u_1 u_5 v_3] \end{aligned}$$

and

$$\begin{aligned} H^7(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1,\dots,5\}} \widetilde{H}^{6-|J|}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \\ &= \widetilde{H}^1(\mathcal{K}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{1,2\}}] \\ &= \mathbb{Z}[-u_3 u_4 u_5 v_1 v_2] \end{aligned}$$

where wee identified the generators in Hochster's formula to generators in the DGA via the given formulas. Denote the generators of H^3 by $x_1, \dots, x_5$ and denote the generators of H^4 by $y_1, \dots, y_5$ and denote the generator of H^7 by z. Then we have the relations $x_1 y_4 = x_2 y_2 = -x_3 y_5 = x_4 y_3 = x_5 y_1 = z$, and 0 otherwise. Thus relabelling the generators give

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \frac{\Lambda_{\mathbb{Z}}[x_1, \dots, x_5, y_1, \dots, y_5]}{(x_i y_i - x_j y_j, x_i y_j, x_i x_j, y_i, y_j \mid 1 \leq i \neq j \leq 5)}$$

where $\deg(x_i) = 3$ and $\deg(y_i) = 4$. Notice that this is precisely the cohomology ring of $(S^3 \times S^4)^{\# 5}$.

There are three descriptions of the cohomology groups:

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

For $L \subseteq J$ such that α_L is a cochain basis in the Hochster formula, we have

$$[\varepsilon(L,J)\chi(J\setminus L, L)] \leftrightarrow [\varepsilon(L,J)u_{J\setminus L}v_L] \leftrightarrow \alpha_L$$

Proposition 8.3.11
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(C^{-*,0}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 1$.
- rank$(H^0(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 0$ for $q \neq 0$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p > q$ or $q > n$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p \geq q > 0$ or $q - p > \dim(\mathcal{K}) + 1$.
- rank$(H^{-1}(C^{-*,4}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- rank$(H^{-(n-\dim(\mathcal{K})+1)}(C^{-*,2n}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = $ rank$(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Proposition 8.3.12
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 1$.
- rank$(H^1(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = $ rank$(H^2(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 0$.
- rank$(H^3(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- rank$(H^{n+\dim(\mathcal{K})+1}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = $ rank$(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Lemma 8.3.13
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then the top cohomology $H^{n+m}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is generated by $[\chi(\{1,\dots,n\}\setminus I, I)]$ for any $I \in \mathcal{K}$ and such that $|I| = n$.

If we use the Koszul complex as the cohomology model for $\mathcal{Z}_\mathcal{K}$, then the top cohomology is generated by $u_{\{1,\dots,n\}\setminus I}v_I$ for any $I \in \mathcal{K}$ such that $|I| = n$.

9 Algebra

9.1 Algebra, Coalgebra and Bialgebra

Let R be a commutative ring. Recall that an R-algebra is a ring A together with a ring homomorphism $R \to A$. Equivalently, it is an R-module with multiplication that is associative and respects the R-module structure.

Proposition 9.1.1
Let R be a commutative ring. Let $A \in \mathbf{Mod}_R$ be an R-module. Let $m : A \otimes_R A \to A$ and $u : R \to A$ be R-module homomorphisms. Then (A, m, u) is an associative algebra with multiplication given by m and identity given $u(1)$ if and only if the following diagrams commute:

$$\begin{tikzcd} (A \otimes_R A) \otimes_R A \arrow[r, "m\otimes 1"] \arrow[d, "\alpha"'] & A \otimes_R A \arrow[r, "m"] & A \\ A \otimes_R (A \otimes_R A) \arrow[r, "1\otimes m"'] & A \otimes_R A \arrow[ur, "m"'] & \end{tikzcd} \qquad \begin{tikzcd} R \otimes_R A \arrow[r, "u\otimes \mathrm{id}_A"] \arrow[dr, "\lambda"'] & A \otimes_R A \arrow[d, "m"] & A \otimes R \arrow[l, "\mathrm{id}_A \otimes u"'] \arrow[dl, "\mu"] \\ & A & \end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Equivalently, the above proposition is the same as saying (A, m, u) is a monoid object in the monoidal category $(\mathbf{Mod}_R, \otimes_R, R)$. Motivated by the categorical description of an R-algebra, we define coalgebras as the formal dual of the above object in the following way.

Definition 9.1.2 (Coalgebras)
Let R be a commutative ring. Let $C \in \mathbf{Mod}_R$ be an R-module. Let $\Delta : C \to C \otimes_R C$ and $\varepsilon : C \to R$ be R-module homomorphisms. We say that (C, Δ, ε) is a coalgebra if the following diagrams commute:

$$\begin{tikzcd} C \arrow[r, "\Delta"] \arrow[dr, "\Delta"'] & C \otimes_R C \arrow[r, "\Delta\otimes 1"] & (C \otimes_R C) \otimes_R C \arrow[d, "\alpha"] \\ & C \otimes_R C \arrow[r, "1\otimes\Delta"'] & C \otimes_R (C \otimes_R C) \end{tikzcd} \qquad \begin{tikzcd} & C \arrow[dl, "\lambda"'] \arrow[d, "\Delta"] \arrow[dr, "\mu"] & \\ R \otimes_R C & C \otimes C \arrow[l, "\varepsilon\otimes \mathrm{id}_A"] \arrow[r, "\mathrm{id}_A\otimes\varepsilon"'] & C \otimes_R R \end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Definition 9.1.3 (Bialgebras)
Let R be a commutative ring. Let $B \in \mathbf{Mod}_R$ be an R-module. Let $m : B \otimes_R B \to B$, $u : R \to B$, $\Delta : B \to B \otimes_R B$ and $\varepsilon : B \to R$ be R-module homomorphisms. We say that B is an R-bialgebra if the following are true.

- (B, m, u) is an R-algebra.
- (B, Δ, ε) is an R-coalgebra.
- Multiplication and comultiplication are compatible:

$$\begin{tikzcd} B \otimes_R B \arrow[r, "m"] \arrow[d, "\Delta\otimes\Delta"'] & B \arrow[r, "\Delta"] & B \otimes_R B \\ B \otimes_R B \otimes_R B \otimes_R B \arrow[rr, "\mathrm{id}\otimes\tau\otimes\mathrm{id}"'] & & B \otimes_R B \otimes_R B \otimes_R B \arrow[u, "m\otimes m"'] \end{tikzcd}$$

  where τ is the natural isomorphism from commutativity of tensor products.
- Multiplication and counit are compatible:

$$\begin{tikzcd} B \otimes_R B \arrow[r, "m"] \arrow[d, "\varepsilon\otimes\varepsilon"'] & B \arrow[d, "\varepsilon"] \\ R \otimes_R R \arrow[r, "\lambda"'] & R \end{tikzcd}$$

  where λ is the natural isomorphism from unitality of tensor products.

- Comultiplication and unit are compatible:

$$\begin{tikzcd} R \arrow[r, "u"] \arrow[d, "\lambda"'] & B \arrow[d, "\Delta"] \\ R \otimes_R R \arrow[r, "\Delta\otimes\Delta"'] & B \otimes_R B \end{tikzcd}$$

where λ is the natural isomorphism from unitality of tensor products.
- Unit and counit are compatible:

$$\begin{tikzcd} R \arrow[rr, "\mathrm{id}"] \arrow[dr, "u"'] & & R \\ & B \arrow[ur, "\varepsilon"'] & \end{tikzcd}$$

9.2 Hopf Algebra

Definition 9.2.1 (Hopf Algebras)
Let R be a commutative ring. Let $(H, m, u, \Delta, \varepsilon)$ be an R-bialgebra. Let $S : H \to H$ be an R-module homomorphism. We say that $(H, m, u, \Delta, \varepsilon, S)$ is a Hopf algebra if the following diagram commutes:

$$\begin{tikzcd} & H \otimes_R H \arrow[rr, "S\otimes\mathrm{id}_H"] & & H \otimes_R H \arrow[dr, "m"] & \\ H \arrow[ur, "\Delta"] \arrow[rr, "\varepsilon"] \arrow[dr, "\Delta"'] & & R \arrow[rr, "u"] & & H \\ & H \otimes_R H \arrow[rr, "\mathrm{id}_H\otimes S"'] & & H \otimes_R H \arrow[ur, "m"'] & \end{tikzcd}$$

9.3 Differential Graded Algebras

Definition 9.3.1 (Differential Graded Algebras)
Let R be a commutative ring. A differential graded R-algebra consists of the following data.
- A $\mathbb{Z}$-graded R-algebra A.
- An R-algebra homomorphism of degree -1 (called homological dgas) or of degree 1 (called cohomological dgas).

such that the following are true.
- Differential: $d \circ d = 0$.
- Graded Leibniz: $d(ab) = (da)b + (-1)^{\deg(a)}a(db)$ for a, b homogeneous elements of A.

The differential property means that any (homological) differential graded algebra decomposes into a chain complex whose differential is given by d.

Definition 9.3.2 (Homomorphisms of Differential Graded Algebras)
Let R be a commutative ring. Let A, B be differential graded algebras. A morphism $\varphi : A \to B$ of differential graded algebras is a graded R-algebra homomorphism.

Lemma 9.3.3
Let R be a commutative ring. Let (A, d) be a cohomological differential graded algebra. Then $H^*(A, d) = \bigoplus_{n\in\mathbb{Z}} H^n(A, d)$ is a graded ring with multiplication defined by

$$[u] \cdot [v] = [uv]$$

for $u \in H^n(A, d)$ and $v \in H^m(A, d)$.

Proof
Notice that by the graded Leibniz rule, we have $d(uv) = (du)v + (-1)^n ud(v) = 0$ since u and v are cocycles. Hence uv is also a cocycle. ■

Definition 9.3.4 (Commutative Differential Graded Algebras)
Let R be a commutative ring. Let A be a differential graded algebra. We say that A is commutative (cdga) if A is graded commutative.

Lemma 9.3.5
Let R be a commutative ring. Let M be an R-module. Let $\varphi : M \to R$ be an R-module homomrphism. Then $K_\bullet(\varphi)$ is a commutative differential graded algebra.

9.4 Models for DGAs

10 Koszul Algebras and Quadratic Algebras

11 Profinite Groups

11.1 Basic Properties

Definition 11.1.1 (Profinite Groups)
Let G be a topological group. We say that G is a profinite group if there exists an inverse system $X : \mathcal{I} \to {}_G\textbf{Top}$ consisting of finite groups with the discrete topology such that

$$G = \lim_{\mathcal{I}} X$$

In particular, this means that G is equipped with the weak topology.

Proposition 11.1.2
Let G be a topological group. Then G is profinite if and only if G is compact and totally disconnected.

Corollary 11.1.3
Let G be a profinite group. Then the following are true.
- G is Haudorff.
- G is locally compact.

Proposition 11.1.4
Let G be a compact topological group. Then G/G_0 is a profinite group.

Proposition 11.1.5
Let G be a profinite group. Let $H \leq G$ be a subgroup. Then the following are true.
- If H is closed, then H is profinite.
- If H is closed and normal, then G/H is profinite.
- H is open if and only if H is closed and $[G : H] < \infty$.

11.2 Profinite Completion

12 Equivariant Algebraic Topology

12.1 G-Homotopy

Definition 12.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G-spaces. Let $f, g : X \to Y$ be G-equivariant maps. A G-homotopy from f to g is a homotopy $H : X \times I \to Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \to Y$$

is G-equivariant map.

12.2 Equivariant CW Complexes

12.3

Definition 12.3.1 (The Homotopy Quotient (Borel Construction))
Let G be a topological group. Let X be a G-space. Define the homotopy quotient of X to be the orbit space

$$EG \times_G X = \frac{EG \times X}{G}$$

where the action of G on $EG \times X$ is given by $g \cdot (e, x) = (g \cdot e, g^{-1} \cdot x)$.

Proposition 12.3.2
Let G be a topological group. Let X be a G-space. Then

$$X \hookrightarrow EG \times_G X \to BG$$

is a fibration.

Lemma 12.3.3
Let G be a topological group. Let X be a G-space. If G acts freely on X, then there is a homotopy equivalence

$$EG \times_G X \simeq \frac{X}{G}$$

induced by the projection map $EG \times_G X \to X$.

13

Definition 13.0.1 (Gauge Group)
Let G be a topological group. Let $p : E \to X$ be a principal G-bundle. Define the gauge group of p to be the set

$$G(p) = \{f : E \to E \mid f \text{ is an isomorphism of } p\}$$

together with composition of functions.

Proposition 13.0.2
Let G be a topological group. Let $p : E \to X$ be a principal G-bundle. Then there is a group isomorphism

$$G(p) \cong \Gamma(\widetilde{p})$$

where $\widetilde{p}$ is the associated principal bundle where G acts on the fibers by the adjoint representation.

Note: The associated principal bundle is the space

$$\frac{E \times G}{\sim}$$

where $(p, g) \sim (ph^{-1}, hgh^{-1})$ for $p \in E$ and $g, h \in G$.

14

There is a one to one bijection between representations of $\pi_1(X)$ and locally constant sheaves (called local systems).

Definition 14.0.1
Let X be a space. Let $x_0 \in X$. Define the functor $\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\mathbf{Sets}$ as follows.
- For $p : \widetilde{X} \to X$ a covering space, define $\text{Fib}_{x_0}(p) = p^{-1}(x_0)$.
- For $f : \widetilde{X}_1 \to \widetilde{X}_2$ a map of covering spaces, define $\text{Fib}_{x_0}(f)$ to be $f|_{p^{-1}(x_0)}$.

Theorem 14.0.2
Let X be a connected and locally simply connected space. Let $x_0 \in X$. Then

$$\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\mathbf{Sets}$$

is fully faithful and essentially surjective.

Theorem 14.0.3 (Classification of Local Systems)
Let X be a space (locally compact, Hausdorff, second countable, locally path connected, semi locally path connected, path connected). Then there is an equivalent of categories

$$\text{Func}(\mathbf{Open}^{\text{op}}, \text{Vect}_{\mathbb{C}}) \simeq \text{Rep}(\pi_1(X))$$

Ref: Galois Groups and Fundamental Groups

Theorem 14.0.4 (Classification Theorem for Flat Connections)
Let M be a connected Riemannian manifold. Let G be a Lie group. Then there is a one to one correspondence

$$\frac{\{(P, A) \mid P \to M \text{ is a principal } G\text{-bundle and } A \text{ is a flat connection}\}}{\sim} \overset{1:1}{\longleftrightarrow} \text{Rep}(\pi_1(X), G)$$

where $(P_1, A_1) \sim (P_2, A_2)$ if there exists bundle isomorphism $\varphi : P_1 \to P_2$ such that $\varphi^* A_2 = A_1$.

Ref: Differential Geometry; Taubes Bijection described in: Geometry of Characteristic classes; Morita

There is a natural bijection between principal bundles and vector bundles by taking the associated bundle. If $G = GL(V)$ for some finite dimensional vector space V over the field $k = \mathbb{R}$ or $\mathbb{C}$, the sheaf of sections of the associated bundle is a locally constant sheaf.

Also: Upgrade to diffeomorphism.

15 Infinite Galois Theory

Proposition 15.0.1
Let F be a field. Let K/F be a Galois extension. Then the following are true.
- The inverse limit of the system $\text{Gal}(L/F)$ for L/F a finite Galois extension is $\text{Gal}(K/F)$.
- $\text{Gal}(K/F)$ is a profinite group.
- The projection maps $\text{Gal}(K/F) \to \text{Gal}(L/F)$ for L/F a finite Galois extension is a surjective homomorphism.
- Let $F < L < K$ be a field extension. Then K/L is a Galois extension and $\text{Gal}(K/L)$ is a closed subgroup of $\text{Gal}(K/F)$.

Theorem 15.0.2 (Krull)

Let F be a field. Let K/F be a field extension. Then there is an inclusion reversing bijection

$$\begin{Bmatrix} \text{Closed subgroups of} \\ \text{Aut}_F(K) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Intermediary subfields} \\ F < L < K \end{Bmatrix}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M, $M \mapsto \text{Aut}_L(K)$.

Proposition 15.0.3
Let F be a field. Let K/F be a Galois extension. Then the above bijection restricts to a bijection

$$\begin{Bmatrix} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Normal Extensions} \\ K < M \end{Bmatrix}$$